

Matgeo-q.2.5.17

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October 10, 2025

Question

Find a unit vector perpendicular to both the vectors **a** and **b**, where:

$$\mathbf{a} = \hat{i} - 7\hat{j} + 7\hat{k} \quad (1)$$

$$\mathbf{b} = 3\hat{i} - 2\hat{j} + 2\hat{k} \quad (2)$$

Solution

Step 1: Setting up the equations

Let the required vector be $\mathbf{n} = n_1\hat{i} + n_2\hat{j} + n_3\hat{k}$. For $\mathbf{n}$ to be perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, their dot products must be zero.

$$\mathbf{a} \cdot \mathbf{n} = 0 \implies n_1 - 7n_2 + 7n_3 = 0 \quad (3)$$

$$\mathbf{b} \cdot \mathbf{n} = 0 \implies 3n_1 - 2n_2 + 2n_3 = 0 \quad (4)$$

This forms a system of linear equations. In matrix form:

$$\begin{pmatrix} 1 & -7 & 7 \\ 3 & -2 & 2 \end{pmatrix} \begin{pmatrix} n_1 \\ n_2 \\ n_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad (5)$$

Step 2: Solving the system using row reduction

We form the augmented matrix and perform row operations.

$$\begin{pmatrix} 1 & -7 & 7 & 0 \\ 3 & -2 & 2 & 0 \end{pmatrix} \quad (6)$$

$$\xrightarrow{R_2 \rightarrow R_2 - 3R_1} \begin{pmatrix} 1 & -7 & 7 & 0 \\ 0 & 19 & -19 & 0 \end{pmatrix} \quad (7)$$

$$\xrightarrow{R_2 \rightarrow \frac{1}{19}R_2} \begin{pmatrix} 1 & -7 & 7 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix} \quad (8)$$

$$\xrightarrow{R_1 \rightarrow R_1 + 7R_2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 \end{pmatrix} \quad (9)$$

Step 3: Finding the general solution

From the row-reduced echelon form (9), we get the equations:

$$n_1 = 0 \quad (10)$$

$$n_2 - n_3 = 0 \implies n_2 = n_3 \quad (11)$$

Let $n_3 = t$, where t is a real number. Then $n_2 = t$. The general solution for a vector perpendicular to **a** and **b** is:

$$\mathbf{n} = \begin{pmatrix} 0 \\ t \\ t \end{pmatrix} = t \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R} \quad (12)$$

Step 4: Finding the unit vector

To find the unit vector, we first find the magnitude of the direction vector

$$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} :$$

$$\left\| \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \right\| = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2} \quad (13)$$

Now, we normalize the vector $\mathbf{n}$ by dividing by its magnitude. The two possible unit vectors are:

$$\hat{n} = \pm \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} \quad (14)$$

$$= \pm \frac{1}{\sqrt{2}} (\hat{j} + \hat{k}) \quad (15)$$

Graphical Representation

Vectors a (red), b (blue), and unit normal $\hat{n}$ (green)

